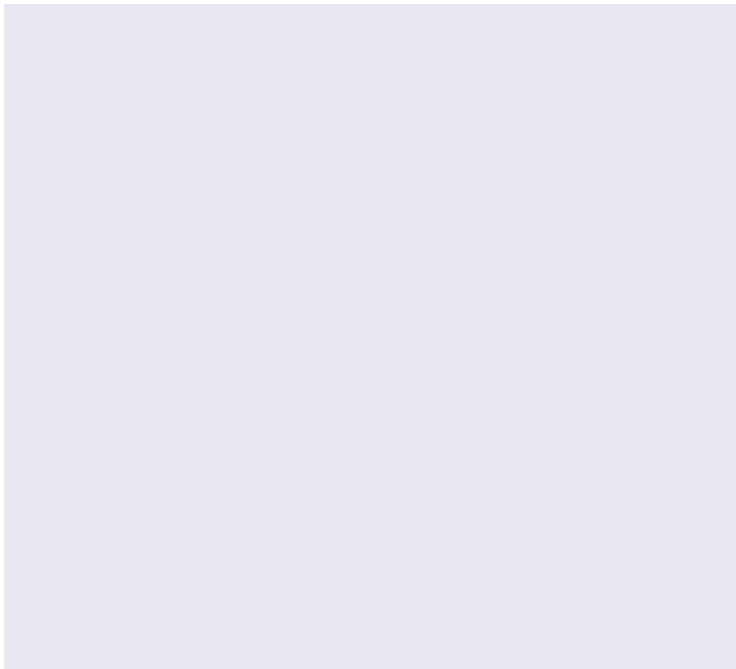



From Clinic to Studio: A Voice Teacher's Guide to Acoustic Voice Analysis

The traditional role of a voice trainer is to assess actors' voice production based on what can be seen and heard (Burke 2014; Sansom 2016). The teacher of voice for actors listens for signs of vocal strain and inefficient or unproductive breath support using visual and auditory perceptual cues. While these methods are tried and true and offer key details about the actor's integration and synthesis of studio training, they tend to operate on a rather subjective level (Burke 2014). In addition, subtle or subclinical changes in voice production may not be readily apparent to the ears or visible to the eyes of the teacher. For these reasons, a thorough understanding of acoustic voice assessment may be a powerful tool to help actors know their own instruments more intimately; allow teachers to connect with students who appreciate technology; objectively track changes in students' vocal performance; and empower teachers to identify phonatory issues at the earliest possible opportunity in order to refer the student for clinical support.



Dramatic intensity and prolonged shouting may cause potential vocal injuries and can therefore limit the actor's vocal flexibility and effectiveness on stage (Titze, Lemke, and Montequin 1997). Emerich et al. (2005) used voice range profiles (VRP) to study the physiologic limits of pitch and loudness ranges in eight professional actors. All the actors exceeded their physiologic VRP baselines in one or more aspects during their performances either in studio or on-stage, despite support from coaches. Most actors were occasionally louder than their dynamic range during acting, and the women actors were seen to consistently lower their pitch below their VRP limits. All these behaviors indicate vocal hyperfunction that can prolong the state of vocal fatigue and increase the potential for injury. Using VRPs as part of vocal pedagogy while training performers could yield better outcomes. Voice trainers can also obtain objective information pre- and post-training that may help refine pedagogical processes and potentially support longitudinal research processes. Interprofessional collaboration and communication with clinicians on a voice care team will undoubtedly be enhanced if the acting voice teacher is familiar with objective information on voice analysis such as acoustic measures.

The pedagogical bond between an actor in training and their voice teacher must include necessary space for support with health and personal challenges that impact their ability to perform (Myers, Manternach, and Titze Cox 2021). A trusted voice teacher or coach is often a key member of an actor's care team when faced with a vocal trauma (or phonotrauma) that could potentially disrupt their career and expression of their artistry. In such cases, a teacher who is fully informed on the implications of a voice disorder through the objective data obtained from instrumental analysis conducted by a clinician can best support the student's voice habilitation. With greater acumen concerning the implications of acoustic voice analysis in particular, acting voice teachers can create a supportive studio setting that will allow the performer's voice to rest, heal, revive, and thrive.

It must be reiterated that voice teachers and coaches know the best practice when presented with a client or student with voice-related signs and symptoms that require intervention: the ethical pedagogue will immediately refer the actor to the appropriate clinician. Crucially, it is key for the voice teacher to recommend a clinician with an expertise in intervention for professional voice users, whether that person is a speech-language pathologist (SLP), laryngologist, otolaryngologist, or ear, nose, and throat specialist (ENT). This is a key distinction: not all SLPs or ENTs specialize in voice for performers. Other key referrals might be to a singing health specialist (SHS), nutritionist, psychologist, or primary care physician.

Great teachers know that referral to a medical intervention team is the best way to ensure long-term vocal health for our clients and students. Kate DeVore's essay "To Refer or Not to Refer" includes helpful language for the voice trainer seeking to find the gentlest way to express the need for clinical referral without alarming the actor or creating additional anxiety. She suggests that minimizing fear around the larger concept of a potential voice disorder will increase confidence in the actor or student and create the circumstances favorable to greatest success: early intervention (DeVore 2001).

With the above considerations in mind, the following is an exploration of acoustic voice analysis for the acting voice teacher or trainer who seeks to enhance their pedagogy with deeper understanding of acoustic parameters of the voice. Such a teacher may also expand their knowledge on voice production through exploring acoustic voice analysis to obtain and track progress on student's vocal performance for pedagogical and



habilitative purposes. Tools such as Praat, Voice Analyst, and VoceVista offer windows into the voice signal. When working with students experiencing mild dysphonia, first-hand knowledge of acoustic voice analysis and perhaps even some personal experience with these tools may offer a scientifically-supported way into thinking and talking about voice disorders. This approach may be less subjective and more effective in collaboration with an interprofessional voice intervention team. As will be discussed, this technology is not appropriate for assessing moderate to severe impairments.

Acoustic Voice Analysis in the Clinical Setting

It is common for actors in training to find themselves with a voice complaint due to an inability to fully integrate new skills for healthy and efficient voice production. With intermediate vocal rest, attention to healthy technique, and improved vocal hygiene, most such cases will resolve themselves quickly and result in an actor more intimately connected to their instrument and committed to healthy use. In the case of an actor with a voice complaint that goes beyond an acute strain, the performing arts voice professional knows referral to a clinician is the logical next step. But what happens next?

In a clinical setting with either a speech language pathologist or physician such as a laryngologist or other voice specialist, it is likely that the voice will be evaluated using an established protocol. Part of this process will involve assessment using an array of instrumental tools that reveal a layer of detail about a pathology that may not be apparent to the ear. However, the actor's subjective insights on their voice problem should also be considered in the evaluation process. Acoustic data offers valuable insights, but acoustic assessment works best when supported by self-perceptual scales (patient-rated), perceptual scales (clinician-rated), aerodynamic measures, and laryngeal imaging (to view structure, movement, and function) (Ma and Yiu 2011, 219). An expert panel assembled by the American Speech-Language Hearing Association (ASHA) emphasizes this in its paper by Patel et al. (2018) on developing consistent industry-wide protocols for instrumental voice assessment:

When evaluating clients with voice disorders, it is preferable, whenever possible, to characterize the impact of the disorder(s) on all of the pertinent mechanisms/dimensions by obtaining complete case histories and performing the following battery of assessments: auditory-perceptual, laryngeal endoscopic imaging, acoustic, aerodynamic, and clients' self-perception of the impact of the voice disorders on their daily function. (887)

What if the actor is diagnosed with a voice disorder or dysphonia? Dysphonia may be broadly defined as an alteration in the production of voice that impairs social and professional communication (Schwartz et al. 2009). Dysphonia is characterized by abnormality in vocal quality, pitch, loudness, or vocal effort that impairs communication or reduces voice-related quality of life. Someone with dysphonia may have decreased or limited ability to interact vocally with others or a self-perceived decrease in physical, emotional, social, or economic status resulting from voice-related dysfunction (Schwartz et al. 2009; Behlau, Madazio, and Oliveira 2015). They will likely be prescribed voice therapy with an SLP along with other treatment modalities as suggested by the voice care team.



Interprofessional collaboration between the voice teacher and the clinical voice care team helps to extend therapeutic techniques into different settings, including structured therapy sessions, exercises for the classroom or studio, and preparation for performances. Depending on voice therapy approaches and collaboration, the intervention may include: vocal hygiene based on lifestyle modifications; techniques to improve coordination among breath, voice, and resonance; simple replacement techniques to address specific symptoms; or a combination of some or all the above to improve vocal endurance, strength, and agility.

When the actor comes off vocal rest and returns to the studio, then what? By the time a student with a diagnosis returns to the classroom, they will likely have undergone a series of tests including a thorough acoustic voice analysis. What are these tools, and what do they measure? And most importantly, how can acoustic measures, combined with the advice of the clinician, help a teacher develop a strategy to support the actor's voice in recovery?

Parameters of Acoustic Voice Analysis

Acoustic voice assessment is an established technique that has been around since the 1960s (Lin, Hornibrook, and Ormond 2012). There are several valuable data points to be obtained from acoustic analysis. Teixeira, Oliveira, and Lopes (2013), Brockmann et al. (2008), de Felipe et al. (2006), and Titze (1994) asserted that the ones most commonly included in voice assessment are fundamental frequency, vocal intensity, jitter, shimmer, and the harmonic-to-noise ratio. Acoustic assessment can be conducted with widely available free computer software such as Praat, WASP, and WaveSurfer, but there are other systems available for purchase. These software systems offer different benefits and can vary greatly in cost (Vaz-Freitas et al. 2018). Some acoustic analysis programs examine a limited aspect of the voice signal, while others make it possible to examine a broader range (Ma and Yiu 2011).

The wide variety of systems available may offer a broad array of options, but there is a potential downside. Titze (1994) and de Felipe et al. (2006) emphasized the need for consistency across the different systems because the stated goal is to obtain more objective quantitative data. Importantly, it has been proven that there is a need to standardize data from the software program being used (de Felipe et al. 2006) and that variance in factors like time interval will negatively impact the accuracy of acoustic measures (Lechien et al. 2021). When analyzing results, it is necessary to compare measures within the same program rather than looking at the same metric across different software systems (Ma and Yiu 2011, 231). Acoustic assessment is a powerful tool to discover aspects of the voice signal that may be imperceptible to the ear, but its benefits can be diminished if users fail to carefully guard accuracy and compatibility of the data by ensuring consistency in settings, readings, microphone placement, software, and environment, among others (Lin, Hornibrook, and Ormond 2012).

While acoustic analysis is a powerful tool, it may not be appropriate for all voices. Titze (1995) created a classification system in which the voice is sorted into three primary types based on the degree of periodicity of the voice signal. Before considering the various voice signal types, it is useful to define perturbation. Perturbation can be considered a "minor disturbance, or a temporary change, from an expected behavior" (Titze 1995, 6).



Perturbations are generally minor enough to be considered “small irregularities” (6). One is observing perturbation in the voice signal when there is a significant variation between the expected and observed outcome in the signal. Perturbation is defined as the cycle-to-cycle variability in a signal and can only be measured from sustained or extracted vowel segments (Baken 1997). Some degree of perturbation may be unavoidable despite producing an almost steady voice. In people with dysphonia or voice disorders, a deviation from normal voice production leads to a significant variation in perturbation measures such as jitter and shimmer

Type I signals have a “nearly periodic” signal and minimal subharmonics; type II signals are mostly periodic but have strong subharmonics; type III signals have “no apparent periodic structure,” and may seem chaotic (Titze 1995, 19). Since the creation of Titze’s system, Sprecher et al. (2010) have suggested the addition of a fourth voice signal type, which distinguishes type III voice signals with a stochastic nature from the main type III (Sprecher et al. 2010). Type IV is a voice signal that is aperiodic and contains a high degree of noise.

In Titze’s system, only type I voices are suited for analysis of perturbation using acoustic measures. For type I voice signals, which are nearly periodic in nature, perturbation analysis using acoustic assessment may be especially useful and likely to render reliable results (Titze 1995, 26). For type III voice signals, perceptual assessments of qualities like breathiness and roughness are generally most useful (Titze 1995, 26), and acoustic analysis of any kind is generally not useful or appropriate (Ma and Yiu 2011, 233). Attempting to conduct acoustic analysis on a type III voice will likely render inaccurate results (Ma and Yiu 2011, 225). However, a type III voice may require the clinician to rely on frequency-based measures such as cepstrum. Cepstrum is a form of spectral voice analysis. It is a mathematical inverse Fourier transformation of the acoustic spectrum which highlights the peaks of the strongest harmonic energies in the acoustic signal (Heman-Ackah, Michael, and Goding 2002; Heman-Ackah et al. 2003). Cepstral measures do not rely on pitch detection algorithms and can be used on highly aperiodic voice signals such as type II and III and on samples of connected speech (Heman-Ackah 2004; Awan, Roy, and Dromey 2009; Peterson et al. 2013). This distinguishes cepstrum measures from time-based perturbation measures such as jitter and shimmer.

With the need for identification of voice type and standardization of equipment and data points established, it is clear that acoustic voice assessment can be a valuable tool offering key insights into several aspects of the voice. While it is possible for a clinician to diagnose a voice disorder using only auditory perceptual analysis, it could render different results depending on the degree of experience of the practitioner (Teixeira, Fernandes, and Alves 2017). With a few key data points obtainable from acoustic analysis, it may be possible to determine whether an actor might benefit from clinical intervention. Because there is a large degree of normative data available, comparing an acoustic voice report to key data points may help distinguish between normal or pathological voices (Teixeira, Oliveira, and Lopes 2013). An acting voice teacher would not make a diagnosis, but knowing how to interpret key acoustic voice analysis measures may help in the making of DeVore’s key question, “To Refer or Not to Refer?” A reminder of her key conclusion: “when in doubt, refer” (DeVore 2001).



Though perceptual observations about voice quality and acoustic analysis may sometimes align, they often diverge (Ma and Yiu 2011, 220). For instance, frequency variation is generally easy to identify perceptually without acoustic measures, but assessment of measures of perturbation such as jitter and shimmer may be more complex (Ma and Yiu 2011, 220). Overall, there should be a relationship between acoustic and perceptual observations; when they diverge markedly, it could be a technical error or a sub-clinical impairment behind a more significant discrepancy. The following is a consideration of what acoustic analysis reveals about fundamental frequency and intensity and aspects of voice quality revealed by measures of jitter, shimmer, and the harmonic-to-noise ratio.

Fundamental Frequency

Although “pitch” is used in common parlance to refer to the perceived highness or lowness of a voice, pitch is a relatively subjective measure. Pitch is the psychological equivalent of a quantitative measure called fundamental frequency. In acoustic analysis, the fundamental frequency (f_0) of a voice is the rate at which the vocal folds vibrate, or the number of vibratory cycles produced by vocal folds per second (Laukka et al. 2008). A vibratory cycle includes one opening and closing of the glottis. While fundamental frequency is related to pitch, they are not exactly the same thing; pitch is how humans psychologically perceive the acoustic phenomenon of f_0 . Fundamental frequency is measured in Hertz (Hz) and can be defined as the number of repeating opening/closing cycles of the glottis (Teixeira, Oliveira, and Lopes 2013). Using acoustic analysis software, one can analyze a recorded voice for attributes like the mean, median, minimum, maximum, and standard deviation of f_0 . As the pitch goes higher, more vibratory cycles are produced per second and vice-versa.

Vocal Intensity

Vocal intensity of a speech signal is measured in units of decibel sound pressure level (dB SPL) (Behrman 2016, 39). Loudness is the perceptual correlate of vocal intensity. When measuring vocal intensity, proper microphone placement is crucial; the distance from mouth to microphone must remain constant to compare samples accurately (Behrman 2016, 174). Vocal intensity is primarily related to the degree of subglottic pressure (Wertzner, Schreiber, and Amaro 2005). Other factors that impact intensity include duration of closing phase of vocal fold vibratory cycle, amplitude of vocal fold vibration, the transglottal airflow difference between subglottis and supraglottis, and the interaction of source (glottal-level) and filter (vocal tract tuning) which may enhance the signal output (Titze and Sundberg 1992; Henrich, Smith, and Wolfe 2011; Brockmann-Bauser, Bohlender, and Mehta 2017).

Jitter

Jitter is the degree of cycle-to-cycle variation in periodicity in an acoustic voice signal (Teixeira, Fernandes, and Alves 2017). Jitter is a short-term assessment of perturbation (Titze 1995, 6). The term “jitter” is derived from the observation that waveforms do not generally display identical periods; because of variation in the f_0 , the non-



periodic waveform appeared to be “jittery” (Titze 1995, 6). Jitter measurements can be higher in individuals exhibiting variations in f_0 such as unstable vocal tone and pitch breaks secondary to poor muscular control over vocal fold vibratory patterns (Teixeira, Fernandes, and Alves 2017). Frequency variation should be seen as an increase in jitter as it denotes the cycle-to-cycle variability in periodicity of an acoustic wave. Researchers generally consider a jitter value between 0.5–1.0% to be typical of normative voices in young adults (Teixeira, Oliveira, and Lopes 2013). Jitter has been shown to increase markedly as loudness decreases (Ma and Yiu 2011, 242).

Shimmer

Shimmer is the cycle-to-cycle variation in amplitude of the voice signal (Titze 1995, 6). Shimmer measurements can be higher in individuals with decreased glottal resistance secondary to increased glottal area due to mass lesions on the vocal folds and/or decreased laryngeal muscular tension due to disruption in neuromuscular function. Shimmer is the magnitude of the variation of the glottal period (Teixeira, Fernandes, and Alves 2017). A variation in glottal period magnitude created by higher airflow passing through the glottis, especially the DC airflow from the posterior glottis, even during the closed phase, causes a higher shimmer reading. A breathy voice may be perceived to have higher shimmer. Likewise, vocal loudness variation is seen as an increase in shimmer as this denotes cycle-to-cycle variability in amplitude of the sound wave. Like jitter, shimmer has been observed to have an inverse relationship with loudness (Ma and Yiu 2011, 242). According to Teixeira, Oliveira, and Lopes (2013), shimmer values of less than 3.8% or 0.350 dB are considered normal in young adults.

Interestingly, Titze (1995, 7) describes the confusion that can emerge from the term shimmer when teachers of singing refer to “a beautiful bell-like vocal quality” this way. “Shimmer” might be a beautiful word, but the quality it is describing in a voice analysis is often very different; it generally refers to a crackling, buzzing quality and can refer to a voice timbre that is “unpleasant and rough” (Titze 1995, 7).

Harmonic to Noise Ratio

Another measure obtainable from acoustic analysis is the HNR, harmonic to noise ratio. HNR reveals information about the degree of periodicity of the voice signal (Teixeira, Fernandes, and Alves 2017). HNR often correlates to perceptual assessments of qualities like roughness (Ma and Yiu 2011, 220) and hoarseness (248). The HNR is the ratio between the periodic component (harmonics) and aperiodic component (noise) of the voice signal. Harmonics arise from periodic components of vocal fold vibration, while noise refers to aperiodic components such as breathiness and is expressed in decibels (dB). The relationship between harmonics and noise indicates the efficiency of speech; a sonorous or normal voice will have a higher HNR, while a low HNR indicates the dominance of noise energy over harmonic energy due to a potential dysphonia (Teixeira, Oliveira, and Lopes 2013).



Implications for the Acting Voice Studio

For the teacher of voice for actors who desires to expand their knowledge of acoustic voice analysis, Praat is a free, open-source program that offers many benefits. Praat can be used to assess acoustic measures in normal voices and in some samples of speakers with voice disorders (Ma and Yiu 2011, 231). The voice professional who has experience working with Praat for functions like spectral analysis, prosodic analysis, and mapping pitch contours may be eager to discover and experiment with additional functionality. Praat's function can be further enhanced by downloading scripts that can automate complex processes and allow the user to customize the software to their individual needs (Ma and Yiu 2011, 232).

Titze's "Summary Statement from the Workshop on Acoustic Voice Analysis" contains valuable information on how to obtain the most accurate results by standardizing equipment and software for acoustic analysis (Titze 1995). This is an indispensable resource for the teacher of voice interested in exploring acoustic assessment.

Praat allows users to analyze the voice signal across several metrics, including f_0 , intensity, formants, and pulses. Ma and Yiu's *Handbook of Voice Assessments* includes a thorough description of Praat's functionality and best practices in Chapter 18: Acoustic Analysis Using Freeware: Praat by Catherine Madill and Patricia McCabe. Both Titze's article and Madill and McCabe's chapter would be excellent starting points for teachers interested in experimenting with these tools and familiarizing themselves with their functionality.

Using Praat, one can examine the acoustic features of the steady state of a vowel using the Voice Report feature under "Show Pulses." Again, it is important to ensure all the appropriate settings for the sample have been selected along with proper measures for voice recording; Ma and Yiu (2011) include great detail on settings in Chapter 18. From the voice report, a user can view measures like the mean and median f_0 , minimum and maximum f_0 (fundamental frequency range), jitter, shimmer, and HNR. A potential studio use for acoustic analysis may be tracking instabilities in pitches and loudness levels during vocal glides at different registers, which can be monitored and compared over time.

Voice range profiles (VRP) or phonetograms provide a thorough description of a student's physiological limits of frequency and intensity. Therefore, they are especially useful for monitoring vocal range in professional voice users (Stemple, Glaze, and Klaben 2020). If a student with dysphonia has difficulties with lower or higher range, it could be reflected in these phonetograms. The progress made during and after voice therapy with an SLP in collaboration with an actor's voice teacher can be tracked using the VRP. These tools can be also useful to monitor the wideness of vocal range over time with practice and performance.

Despite the fact that Praat and other software have a bit of a learning curve, Lin, Hornibrook, and Ormond (2012) found that even iPhone recordings of the voice signal proved useful for assessing certain acoustic features. As technology becomes increasingly ubiquitous and accessible, it stands to reason that the modern voice teacher might desire to be better informed about the potential of these tools, their



potential application to the learning environment, and how to use them to obtain essential information about the voice signal that may be not readily apparent to the “naked ear.” A responsible voice teacher will make an immediate referral to a clinician when that data indicates something outside the expected range. Frequency-based perturbation measures such as cepstral analysis can also be measured with the latest version of Praat for students with significant dysphonia as described by an SLP or other clinician.

Some acting voice pedagogues may wonder if integration of acoustic measures may be a threat to their role in the classroom (Sansom 2016). However, it is important to acknowledge that voice technology cannot replace the teacher’s work in vocal development (Miller 1996, 304). These tools can only support the well-trained ear of the voice teacher, but they have great potential for offering teacher and student to track vocal development using objective measures (Miller 1996, 304).

For the voice teacher looking to increase their understanding of acoustic measures, experimenting with acoustic analysis using software may prove invaluable. Taking care to maintain all the proper settings is key, but once the user attains a degree of comfort with the software, it can prove rather intuitive. After exploring the functionality of this acoustic assessment outside the studio or classroom, the teacher may desire to introduce the students to Praat or another program. It could be that in learning about acoustic analysis of voice, a voice teacher or student may discover a reading that appears to be outside the normative values. This could indicate the potential of a voice disorder that may require a clinical referral. Sometimes what is revealed in the acoustic assessment may offer a greater degree of detail or precision than what is possible with perceptual assessment alone. Voice therapy with an SLP combined with performance training with an acting voice teacher may work best when all parties can observe and communicate about objective changes in acoustic information. By familiarizing oneself with these tools and using them to support student learning about phonation, the voice teacher may even help a student to discover an issue that needs attention before it becomes a voice disorder that is difficult to navigate and heal. This could prevent the potential worsening of an existing disorder and help to improve overall vocal health outcomes.

Acoustic analysis can help clinicians and acting voice teachers function as more effective members of the intervention team and hopefully, improve outcomes for performers with voice disorders. Using data obtained from acoustic measures, teachers and clinicians can work in collaboration with the student to develop an intervention plan including schedules, practice, exercise regimens, lifestyle choices, education on vocal functioning and health, and other vocal and physical activities that will support the actor’s voice habilitation.

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